

ELECTRICAL INSTALLATION

An electrical installation course provides comprehensive training in residential, commercial, and industrial wiring, emphasizing safety, practical skills, and regulatory compliance (e.g. IEEE/local codes). Core modules include electrical principles, blueprint reading, conduit bending, circuit testing, maintenance, and solar PV technology, typically spanning from 4-months certificate programs to 2-year diplomas.

Core Curriculum Components

The curriculum is designed to produce competent, safe technicians through theoretical and practical training:

- **Electrical Fundamentals:** Basic electrical principles (voltage, current, resistance), Ohm's Law, power calculations, and basic electronics.
- **Safety and Regulations:** Occupational Health & Safety (OHS), grounding, and strict adherence to electrical codes (e.g. PEC standards)
- **Wiring and Installation:**
 - **Domestic & Commercial:** Installation of lighting, switches, sockets, consumer units, and distribution boards.
 - **Industrial:** Motor controls, 3-phase systems, transformers, and specialized machinery.
 - **Conduit and Cabling:** Conduit bending (PVC/Metallic), conduit installation, cable pulling, and cable tray installation
- **Technical drawing and Layout:** Reading, interpreting, and drawing electrical schematics, wiring diagrams, and building plans.
- **Testing and Troubleshooting:** Inspection, testing of insulation resistance, earth fault loop impedance, and fault diagnosis.
- **Specialized Areas:** Solar photovoltaic (PV) installation, HVAC systems, and environmental safety in damp areas.

Typical Course Structure

- **Artisan Level (Level 3):** Basic wiring and safety (Approximately 1 term/3-4 months)
- **Certificate Level (Level 4/5):** Detailed installation, planning, and design (approx. 2-4 terms/1 year).
- **Diploma Level (Level 6):** Advanced design, management, and complex systems (up to 2 years)

Practical Skills Assessment

Practical competency is a core requirement involvement:

- Real-world wiring projects in workshops.
- Continuous assessment of wiring quality
- Performance of specific tasks such as installing conduit and pulling cables.
- Final installation projects.

Important: If needed to narrow this down, details can be provided on:

- Specific topics within industrial electrical training.
- National certification in Kenya
- Common careers and salaries after completing course.